

Aadithya AR

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SUMMARY

Computer Science (AI & ML) undergraduate with hands-on project experience across the applied ML lifecycle — model development, evaluation, and deployment — spanning clinical ML, multimodal systems, and retrieval-augmented generation. Built and deployed several end-to-end ML systems using FastAPI and AWS, with a focus on measurable performance and explainability.

EDUCATION

Global Academy of Technology

B.E. in Computer Science Engineering (AI & ML) | CGPA: 8.47/10.0

Bangalore, India

Oct. 2023 – Jul 2027

- Coursework:** Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, Machine Learning, Deep Learning

EXPERIENCE

Machine Learning / AI Intern (Agentic AI)

Feb 2026 – Jun 2026

Innomatics Research Labs

Remote

- Built a FastAPI backend for ML model deployment with Pydantic validation and pytest coverage; deployed on AWS EC2 with environment-based config, achieving p99 response under 180ms.
- Designed a reusable NLP preprocessing library (tokenization, stemming, lemmatization, TF-IDF vectorization) used across 3+ downstream ML tasks — compatible with HuggingFace and Scikit-learn workflows.
- Built ML data pipelines using LangChain and Python — prompt engineering, LLM chaining, and document-grounded QA; delivered 2 end-to-end pipelines with sub-3s inference latency.

PROJECTS

Parkinson's Voice Detection — Clinical ML Rebuild

Apr 2026

Python, PyTorch, wav2vec2-XLS-R-300M, praat-parselmouth, Optuna, XGBoost, LightGBM, Flask

[GitHub](#)

- Rebuilt the ML data pipeline and feature extractor — fixed 7 silent failures (aliased MDVP calls, NoneType hardcodes, subject-level data leakage); extended feature set from 22 to 56 dimensions (CPP, MFCCs, formants).
- Built a model serving pipeline using wav2vec2-XLS-R-300M + Logistic Regression: CV AUC 0.972 ± 0.034 , Subject AUC 0.996, 94.25% accuracy on 831 recordings / 61 subjects; ran 75 Optuna hyperparameter tuning trials.
- Preparing findings for submission to a Scopus-indexed conference (targeting IEEE INDICON 2026).

MoodScript — Multimodal Emotion-Aware Journaling

Jun 2026

Python, PyTorch, FastAPI, React, PostgreSQL, HuggingFace, Groq (LLaMA-3.3-70B)

[GitHub](#)

- Built a multimodal journaling system that fuses a text model (DistilRoBERTa) and a facial-expression model (ViT) via calibrated, log-linear fusion; deployed as four independent services (React frontend, FastAPI orchestrator, text/face services) across Cloud Run and Render.
- Diagnosed a calibration mismatch between the two models (Expected Calibration Error 0.235 text vs 0.029 face) that made naive fusion underperform the stronger single model; the calibrated fusion rule raised accuracy to 92.03% / 92.68% across two benchmarks, validated with McNemar's test.
- Preparing findings for submission as an IEEE conference paper on the multimodal fusion methodology.

Adaptive RAG System with XAI + MLOps

May 2026

Python, Mistral-7B via Ollama, ChromaDB, BGE Embeddings, LangChain, RAGAS, W&B, Gradio

[GitHub](#)

- Built a document QA system over 50 ArXiv ML papers with adaptive query routing — switches between BM25 and dense retrieval (70/30 hybrid); 1,206 chunks embedded with model deployment via Ollama inference layer.
- Achieved RAGAS Context Precision 1.000 and Answer Relevancy 0.734; added explainability using SHAP for source attribution and W&B experiment tracking.

TECHNICAL SKILLS

Languages: Python (primary), Java, SQL, JavaScript

Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems

ML / NLP: PyTorch, Scikit-learn, NumPy, Pandas, HuggingFace (wav2vec2, BGE), XGBoost, LightGBM, RAGAS, W&B, Explainable AI (XAI), Feature Engineering, Model Evaluation, Cross-Validation, Model Deployment

GenAI & RAG: LangChain, Prompt Engineering, RAG, BM25, ChromaDB, Ollama

Backend & Cloud: FastAPI, Flask, RESTful APIs, AWS (EC2, S3), Docker, Git, CI/CD

CERTIFICATIONS & ACHIEVEMENTS

2nd Place — GAT Departmental Project Competition (15 teams, 60 students)

Nov 2025

Top 20 of 1,500+ Teams — Google Cloud Gen AI Academy APAC, Cohort 2 (Cohorts 1 & 2 completed, both certified)

2026

IBM Datathon Participant — National Level, 2 consecutive editions

Jan 2026

Machine Learning Foundation — Infosys Springboard

Nov 2025